

FROM DESCRIPTIONS TO SOURCEABLE PARTS: CATALOG-GROUNDED RETRIEVAL FOR MECHANICAL DESIGN

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ABSTRACT

AI engineering agents are beginning to translate textual requirements into mechanical part and component descriptions. To support procurement, bill-of-material generation, and CAD insertion, these descriptions must be grounded in existing catalog parts with exact part numbers. Product-data APIs can attach live links, prices, datasheets, and CAD files after identifiers are known; catalog-grounded part-number retrieval is the upstream step that makes these APIs usable by design agents. We present an open-source Model Context Protocol server for headless navigation of McMaster-Carr that converts sufficiently detailed component descriptions into matching catalog part numbers. The system renders live catalog pages, extracts visible product-table schemas, uses a language model to map requested constraints onto live field names and values, and then performs deterministic row filtering. The output is the set of catalog parts that satisfy the grounded description. In final benchmark runs, the system recovered 250/250 exact part numbers across 25 catalog categories, preserved multi-part alternatives in 50/50 underspecified cases, and returned no part in 25/25 nonexistent cases. By making part sourcing callable by agents, the method enables downstream workflows in which assemblies are crafted from off-the-shelf parts and automatically linked to product data, prices, datasheets, CAD, and bills of materials.

Keywords: AI engineering agents, design automation, part sourcing, model context protocol, catalog grounding, bill of materials, CAD retrieval

1. INTRODUCTION

AI engineering agents are increasingly used to generate mechanical part requirements, component selections, fixtures, subsystems, and assembly concepts from text. These outputs become actionable when they can be connected to existing parts that can be purchased, inspected, trusted, and inserted into CAD. Procurement, bill-of-materials generation, and CAD retrieval all depend on exact catalog identifiers, while generated requirements are typically expressed as natural-language component descriptions. This paper addresses the catalog-grounding step

between language-level engineering intent and sourceable mechanical parts.

McMaster-Carr is a widely used mechanical component catalog, and its Product Information API retrieves product information, live links, prices, images, CAD, and datasheets once a subscribed part number is known [1]. This identifier-centric interface is well suited to downstream product-data retrieval. Design agents, however, require an upstream capability: converting a textual component requirement into the catalog part numbers that satisfy it. A part-sourcing tool with this capability can connect generated component descriptions to live product data, CAD files, and bill-of-material entries.

Catalog-grounded part retrieval is challenging because industrial catalogs exhibit heterogeneous product schemas. A screw may be distinguished by material, thread size, length, drive style, package quantity, and finish. A switch may be distinguished by circuit, terminal count, maintained or momentary action, panel thickness, voltage, and terminal style. Conveyor rollers, springs, hinges, tapes, bearings, and filters each expose different discriminating fields. General retrieval therefore benefits from observing the live schema of the current catalog page and returning the complete set of rows that satisfy the description.

We present a dynamic catalog-grounding method exposed through a Model Context Protocol (MCP) server. Given a textual mechanical component description, the server returns the set of part numbers whose live catalog rows satisfy the grounded constraints. The system derives catalog search queries internally, renders live catalog pages, extracts product rows and visible column names, summarizes the live schema, aligns requested constraints with live fields and values, and filters rows programmatically. The language model serves as a schema-alignment component that produces grounded field-value matchers for the currently rendered catalog page. Once part numbers are recovered, existing product-data APIs can automatically attach live links, price information, datasheets, and CAD files to generated component or assembly descriptions.

The contribution of this work is an open-source MCP server that takes a textual mechanical component description and returns

the matching McMaster-Carr part numbers. The server packages headless rendered-page navigation, runtime product-schema extraction, language-model schema alignment, and deterministic row filtering behind a tool interface that can be called by engineering agents.

2. RELATED WORK AND BACKGROUND

Our review focuses on three bodies of related work: tool protocols for connecting language models to external systems, McMaster-Carr product-data interfaces, and prior McMaster-Carr agent tooling.

2.1. Model Context Protocol for Engineering Tools

The Model Context Protocol standardizes how language-model applications connect to external context and tools [2]. MCP servers can expose resources, prompts, and tools; tools are the primitive used when a model needs to execute an action such as querying a database, calling an API, or driving an external system [3]. Our implementation uses MCP as the integration layer so that the same part-sourcing capability can be called by any compatible agent host.

MCP is useful here because part sourcing is not simply text generation. It requires a persistent browser session, rendered-page inspection, navigation, timeouts, and a structured result contract. These actions are better exposed as tools than hidden inside a prompt.

2.2. McMaster-Carr Product Data

McMaster-Carr’s Product Information API is available to approved customers and includes requests for product information, price, image retrieval, CAD retrieval, and datasheet retrieval once a subscribed part number is known [1]. This makes part-number discovery the critical upstream step. Our system is designed to fill that step and then hand part numbers to an approved product-data pipeline.

McMaster-Carr’s web search is effective for human catalog browsing: a user can enter keywords, inspect category pages, compare table rows, apply filters, and choose a part number. An autonomous engineering agent needs a different interface. Given a product description, the agent needs a structured result containing the part numbers that satisfy the stated requirements. The proposed MCP server is therefore better suited to agentic sourcing because it converts rendered search and catalog pages into a machine-readable matching set rather than leaving the selection step to a human reader.

2.3. Existing McMaster-Carr Agent Tooling

A prior public repository, `mjbrown/mcmaster-agent`, is a useful MCP wrapper around McMaster-Carr search and product lookup [4]. If the user already has a part number, its product tool returns the product name, URL, and specification fields. If the user has a keyword query, its search tool returns candidate product hits with part numbers, names, and URLs. It also includes a URL tool for constructing product or search links. In other words, it helps an agent look up a known identifier or browse search results.

The problem addressed in this paper is one step earlier in the sourcing workflow. A design agent often starts with a product description, not a part number: for example, a detailed requirement for a screw, hinge, bearing, filter, or other catalog component. The contribution here is a tool that takes that description and returns the McMaster-Carr part numbers whose catalog rows match it. This is better suited to automated sourcing because the tool performs the selection step itself: it navigates rendered catalog pages, reads the live table schema, maps the requested constraints to the current fields, and filters the rows. The output can then be handed directly to product-data, CAD, datasheet, and BOM retrieval tools.

3. SYSTEM DESIGN

3.1. Part-Number Resolution Interface

The package exposes a Python MCP server whose central tool, `mcmaster_find_exact_part`, resolves a textual mechanical component description to the set of McMaster-Carr part numbers that match it. This is the interface studied in the paper. The input is a natural-language requirement such as a detailed description of a screw, hinge, switch, bearing, filter, or other catalog component. The output is a structured match set containing the recovered part numbers and supporting evidence from the rendered catalog rows.

This contract is intentionally narrower than general catalog browsing. The server may use internal navigation, extraction, and retry operations to reach the result, but the research object is the description-to-part-number resolver. The implementation is packaged as `mcmaster-navigator-mcp`, requires Python 3.11 or newer, and uses SeleniumBase UC mode with headless Chrome. Model credentials are supplied at runtime rather than embedded in the package.

3.2. Headless Browser and Rendered-Page Extraction

McMaster-Carr pages are rendered web pages with product tables, category links, filters, and product detail pages. The navigator opens search URLs, auto-drills through category links when needed, and snapshots the rendered HTML. The extractor returns part numbers, product rows, row attributes, evidence text, navigable links, and schema objects containing filters, table columns, and part-number rows.

The browser is isolated by default with a temporary Chrome profile. The server serializes browser work through a single worker so browser state remains coherent across navigation calls. Browser logs are kept off MCP protocol stdout.

3.3. Dynamic Exact-Part Resolution

Figure 1 summarizes the exact-part resolver. The resolver derives catalog search roots and literal constraints from the description, searches the generated roots headlessly, extracts rows from product hits and schema tables, summarizes the live schema, asks the language model to map requested constraints to live fields and accepted values, and applies the resulting matchers with Python row filtering. If filtering conflicts with the live schema, the language model repairs the mapping from the extracted field/value summary. If one row remains, the resolver runs a final strict

Text requirement → derived search roots and constraints → headless rendered pages → live rows and schema fields → LLM field/value matchers → deterministic row filtering → matching part-number set.

FIGURE 1: Dynamic exact-part resolution. The language model maps constraints to live page fields; Python filtering determines the returned set of matching part numbers.

verification check against that row. If required constraints are missing or contradicted, the row is removed from the result set.

This design lets the system generalize across product families because the catalog schema is discovered at runtime. There is no static list of all possible attributes for screws, springs, switches, hinges, bearings, hose fittings, raw materials, or other categories.

3.4. Assembly-Level Use Case

The core problem is part sourcing. Assembly workflows become possible by treating each off-the-shelf line item as an independent exact-part resolution problem. A component requirement is sent to `mcmaster_find_exact_part`, and the returned matching set is carried forward. Downstream systems can retrieve product data, live links, price, image, CAD, and datasheet information for returned part numbers, preserve alternatives for later selection, or revise line items whose descriptions do not match any catalog row.

This line-item independence allows parallel sourcing agents. Compatibility checking can be layered above the resolver, but the core tool intentionally solves the narrower and measurable problem of finding the catalog part numbers that match stated requirements.

4. EVALUATION METHODOLOGY

The evaluation measures whether the resolver returns the correct catalog match set from text rather than a plausible ranked search result. We therefore test three regimes: descriptions that should identify one known catalog part, descriptions that intentionally leave several valid alternatives, and descriptions whose required constraints cannot be satisfied.

4.1. Exact Part Recovery

The exact recovery benchmark evaluates precision when the description is sufficiently specific to identify one catalog row. We begin with real McMaster-Carr part numbers, extract the visible catalog attributes for each part from rendered product pages or table rows, transform those attributes into a textual component description, remove the part number, and ask the resolver to recover the matching part-number set. A case is counted correct only when the returned set contains exactly one part number and that part number is the held-out target.

The final exact-recovery run used 250 cases sampled across 25 catalog categories, summarized in Fig. 3.

4.2. Multi-Match Part Recovery

The multi-match benchmark evaluates whether the resolver preserves alternatives for under-specified descriptions rather than forcing a single choice. The 50 cases cover two common forms

Tool call: `mcmaster_find_exact_part`
Input: 316 stainless steel polished surface-mount hinge with holes; nonremovable pin; 2-3/8 in by 1-3/16 in leaves; 0.315 in leaf thickness; 180° range; four holes; fasteners not included.
Returned match set: {1488A1}
Held-out target: 1488A1

FIGURE 2: Representative exact-recovery case. The prompt omits the part number.

Adhesives	Bearings	Building & Grounds	Conveying	Electrical & Lighting
Fabricating	Fastening & Joining	Filtering	Furniture & Storage	Hand Tools
Hardware	Heating & Cooling	Lab Supplies	Linear Motion	Lubricating
Measuring & Inspecting	Motors	Pipe/Tubing/Hose/Fittings	Plumbing & Janitorial	Pneumatics
Power Transmission	Pulling & Lifting	Raw Materials	Shipping	Suspending

25 catalog families: all sampled exact-recovery descriptions resolved to the held-out part number.

FIGURE 3: Exact-recovery coverage across catalog families. The grid reports the diversity of product families represented in the 250-case exact-recovery run.

of ambiguity. In 25 controlled cases, we start from successful exact cases, collect the live schema rows around the target, and remove one discriminator from the description. This creates a new description with a known matching set, usually two to four parts, computed from the same rendered catalog rows. The other 25 cases use broader descriptions such as “double sided tape,” “ball bearing,” “door hinges,” and “toggle switch.” These test the same behavior at larger cardinalities. A multi-match case passes when the resolver returns the matching set for controlled cases, or returns multiple matching part numbers for broad cases, without inventing a single best answer.

4.3. No-Match Part Descriptions

The no-match benchmark evaluates whether the resolver can return an empty set when the description contains requirements that no visible McMaster-Carr row satisfies. These cases start from exact descriptions and add impossible required constraints such as an unobtainable material, impossible color, or impossible size. A case passes only when the resolver returns no part numbers. This tests the failure mode most important for automated sourcing: the tool should report that no catalog match was found rather than selecting a nearby but incorrect part.

4.4. Parallel Execution for Assemblies

5. EXPERIMENTAL RESULTS

5.1. Exact Part Recovery Results

The exact-recovery benchmark returned exactly one part number for all 250 cases, and that part number was the held-out target in every case. This corresponds to 250/250 exact recovery, or 100% accuracy. The run used an average of 7,449.96 recorded LLM tokens per case and an average wall-clock time of 71.038 s per case.

The absence of misses across these categories is evidence that runtime schema extraction can handle substantially different product families. It is not proof of complete coverage of every McMaster-Carr page or every possible human description.

Exact: 250/250 one held-out part recovered
Multi-match: 50/50 alternatives preserved
No match: 25/25 empty set returned

All completed benchmark regimes returned the expected match-set behavior.

FIGURE 4: Benchmark outcome summary. The resolver returns one part for sufficiently specific descriptions, preserves alternatives for under-specified descriptions, and returns no parts for impossible descriptions.

The strongest defensible claim is that dynamic schema extraction and deterministic filtering can recover exact part numbers across diverse catalog families when descriptions contain sufficient identifying information.

5.2. Multi-Match Part Recovery Results

The multi-match benchmark passed all 50 under-specified cases, for a 100% multi-match pass rate. For controlled cases, the resolver returned the expected matching set. For broad category-level descriptions, it returned multiple part numbers rather than collapsing the request to a single guess. These results are reported together because they test the same contract: when a description admits more than one catalog row, the returned object should preserve the alternatives. Across the combined multi-match set, the resolver used an average of 5,573.02 recorded LLM tokens per case and an average wall-clock time of 57.657 s per case.

5.3. No-Match Part Description Results

The no-match benchmark returned empty part-number sets for all 25 cases, for a 100% no-match pass rate. This indicates that, for the tested impossible descriptions, the resolver did not force a nearest-neighbor part when the required constraints could not be satisfied. The run used an average of 10,731.72 recorded LLM tokens per case and an average wall-clock time of 75.348 s per case.

5.4. Summary of Experimental Results

Figure 4 summarizes the completed benchmark runs. The overall reported benchmark set contains 325 cases: 250 exact-recovery cases, 50 multi-match cases, and 25 no-match cases. All 325 returned the expected match-set behavior. Figures 5 and 6 compare the recorded token and time costs across benchmark categories; the average bar is the unweighted mean across exact, multi-match, and no-match categories.

5.5. Assembly Demonstration

The assembly demonstration evaluates the resolver as a line-item sourcing component. Each off-the-shelf component description in an assembly is sent independently to `mcmaster_find_exact_part`; the returned match set is then available to product-data tools that attach live links, prices, datasheets,

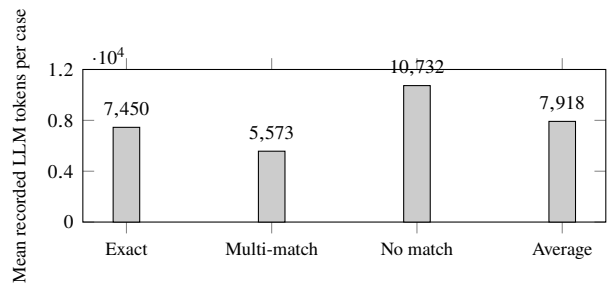


FIGURE 5: Mean recorded LLM tokens by benchmark category. The average bar is the unweighted mean of the three category means.

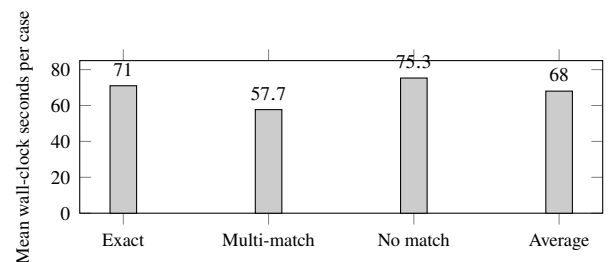


FIGURE 6: Mean wall-clock time by benchmark category. The average bar is the unweighted mean of the three category means.

CAD files, and bill-of-material entries. The demonstration section will report the final sourced assembly, the recovered part-number sets for each line item, and the downstream artifacts attached to the bill of materials.

6. DISCUSSION

6.1. Why Dynamic Schemas Matter

The benchmark covers product families whose discriminating attributes differ substantially. A static parser with a fixed list of fields would need to know domain-specific names and value formats for hinges, switches, bearings, adhesives, filters, raw materials, pneumatic cylinders, motors, and conveyor rollers. The dynamic approach avoids that requirement. The system only needs a general row representation and a way to align user constraints with fields currently visible on the page.

6.2. Why Returning the Set Matters

Procurement workflows need faithful part sets. Returning a single best guess for “ball bearing” is worse than returning the matching catalog rows, because the description does not specify shaft diameter, housing diameter, width, seal type, load capacity, or material. The multi-match and zero-match experiments show that the resolver can return all matching parts, or none, without false precision.

6.3. Integration with Product-Data APIs

Once part numbers are known, product-data retrieval is a conventional API task. The McMaster-Carr Product Information API includes endpoints for product information, price, images, CAD, and datasheets [1]. This separation lets the proposed MCP server remain focused on the hard upstream retrieval problem: finding the part numbers from text and navigation.

7. LIMITATIONS AND FUTURE WORK

The benchmark descriptions were generated from rendered catalog rows and product pages. They are specific and often include labels such as family, group, selected option, and attribute names. This is a valid test of schema grounding and exact retrieval, but it is not the same as testing noisy human design notes.

The system depends on rendered website structure. If McMaster-Carr changes page layout, table markup, anti-automation behavior, or part-number presentation, extraction may require maintenance.

The system is not affiliated with McMaster-Carr and should be used in accordance with applicable terms and approved API

access. The official product information API requires approved customer access and subscription to product information.

Latency is dominated by headless browsing. The mean exact-recovery latency in the final run was 71.038 seconds per case. Parallel line-item sourcing can reduce assembly-level wall-clock time, but single-line resolution remains slow relative to a pure API call.

Future work should evaluate noisier human-written requirements, demonstrate full assembly-level sourcing with compatibility checks across line items, and measure robustness over time as catalog pages change.

8. CONCLUSION

This work turns a practical part-sourcing bottleneck into a measurable tool interface. A design agent can describe a required component, but a downstream product information API needs a part number. The proposed MCP server bridges that gap by headlessly navigating McMaster-Carr, extracting live product schemas, grounding textual constraints to live fields, and filtering rows deterministically. In final benchmark runs, the system recovered 250/250 exact part numbers across 25 catalog categories, preserved multi-part alternatives when requirements were under-specified, and returned no part when requirements could not be satisfied. Reliable part sourcing can let agents compose off-the-shelf components into larger designs while automatically attaching BOM entries, CAD, datasheets, prices, live links, and product data.

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